Test Plan — VWO-49: [VWO] Add Passkey and SSO Login Options to VWO Login Page

Company context: The supplied company profile describes VWO as a Wingify product focused on website A/B testing. **Status: DRAFT — pending human review.** This plan is not approved until a QA owner signs off.

Ticket detail	Value
Issue	VWO-49
Issue status	To Do
Priority	Medium in issue metadata; “High” in the description (conflict to resolve)
Epic	Authentication Enhancements
Components / fix version / sprint	Not specified
Out of scope	Passkey registration; SSO provider configuration

1. Scope & Objectives

- **In scope:** Login-page visibility and placement of SSO and Passkey options; successful and unsuccessful SSO authentication; successful and unsuccessful Passkey authentication; no-registered-passkey and unsupported-environment handling; configured identity-provider compatibility; security, accessibility, responsive behavior, and relevant login regression checks.
- **Out of scope:** Passkey registration and SSO identity-provider configuration. Test setup may use existing configured providers and previously registered passkeys, but changing those configurations is not covered.
- **Objective:** Verify that eligible users can choose either new login method, reach the dashboard on successful authentication, receive an agreed safe and usable recovery/error path on failure, and do not expose sensitive authentication data. Confirm behavior across the product’s supported browser/device and configured identity-provider matrix.
- **Requirement sources:** VWO-49 acceptance criteria and listed test scenarios. No separate design, provider matrix, or detailed security specification was included in the ticket export.

Requirement checklist review

Checklist area	Status	Finding
User goal / story	✅ Present	User wants secure and convenient login using Passkey or SSO.
Testable acceptance criteria	⚠️ Ambiguous	Successful outcomes are observable; exact messages, fallback behavior, and some protocol/security expectations are not.
Happy paths	✅ Present	SSO success redirects to dashboard; verified Passkey logs in and redirects.
Negative / error paths	⚠️ Ambiguous	Error categories are named, but expected content, recovery, retry, and cancellation behavior are not defined.
Boundaries / empty states	❌ Missing	No details for missing SSO identifiers, cancelled prompts, timeout, repeated failures, or other boundary conditions.
State transitions	⚠️ Ambiguous	Login-to-provider/authenticator-to-dashboard is implied; intermediate states and return/cancel paths are unspecified.

Checklist area	Status	Finding
Test data / preconditions	❌ Missing	No test accounts, tenant setup, registered passkey profile, or invalid credential data identified.
Environment / configuration	❌ Missing	No staging URL, feature flag, environment configuration, or supported browser/device list specified.
Dependencies / integrations	⚠️ Ambiguous	OAuth/SAML and configured identity providers are mentioned, but provider names, versions, and test ownership are absent.
Performance	❌ Missing	No response-time or availability expectation for either authentication flow.
Security / authorization	⚠️ Ambiguous	Secure protocols and no sensitive data exposure are required, but protocol details, role coverage, and logging scope are unspecified.
Accessibility	⚠️ Ambiguous	Accessibility is listed as a test scenario but no conformance target or expected keyboard/screen-reader behavior is defined.
Internationalization	❌ Missing	No supported locales or translated-message expectations specified.
Audit / observability	⚠️ Ambiguous	Network-log exposure is prohibited; required audit events and safe diagnostic logging are not defined.
Regression impact	⚠️ Ambiguous	Existing login methods are present in context, but no explicit regression surface or expected behavior is listed.
Backward compatibility / migration	❌ Missing	No expected behavior for existing accounts, tenants, or authentication configuration changes.
Browser / responsive matrix	⚠️ Ambiguous	Passkeys are limited to supported browsers/devices and responsive behavior is mentioned, but the support matrix and breakpoints are absent.
Rollback / feature flag	❌ Missing	No flag or rollback behavior specified.
Terminology / clarity	⚠️ Ambiguous	“Valid SSO credentials” may not describe all federated SSO flows; “fallback or guidance” leaves the expected Passkey outcome open.
Design / mockups	❌ Missing	No design link or approved copy for the login controls and error states is provided.

2. Gaps & Questions for the author

#	Area	Finding	Question to author
1	Priority	⚠️ Issue metadata says Medium; description says High.	Which priority should drive release and test triage?
2	SSO flow	⚠️ “Valid/invalid SSO credentials” is underspecified for federated identity.	Which identity providers and protocols are in scope, and what are the expected success, rejection, account-linking, cancellation, timeout, and recovery outcomes for each?
3	Passkey failures	⚠️ “Fallback or guidance” and “authentication failure” do not define exact behavior.	What should users see/do for no registered passkey, user cancellation, timeout, authenticator rejection, and invalid/expired challenge? Is password login the intended fallback?
4	Error copy	⚠️ Expected message and page state are not defined in the AC.	What approved message, retry behavior, and destination should each failure use? The ticket comment’s “Invalid credentials” is an execution expectation, not an explicit AC.
5	Test setup	❌ Test identities and prerequisites are absent.	Can QA get accounts for successful/failed SSO per provider, accounts with/without registered passkeys, and access to provider test tenants?
6	Compatibility	⚠️ No browser, OS, device, or provider support matrix.	Which exact desktop/mobile browsers, operating systems, authenticators, and identity-provider configurations must pass?
7	Security	⚠️ Protocol names and no-exposure rule are broad.	Which OAuth/SAML/WebAuthn versions/configurations are expected, which roles/tenants must be covered, and what specific UI, browser storage, URL, and log fields are considered sensitive?

#	Area	Finding	Question to author
8	Quality attributes	❌ Performance and localization targets absent; accessibility target unclear.	What performance thresholds, supported locales, and accessibility standard/level should be used?
9	Existing login / rollout	⚠️ Regression and rollback expectations are not stated.	Which existing login methods and account states must be regression-tested, and is there a feature flag or rollback procedure?
10	Design / ownership	❌ No mockup, component owner, fix version, or sprint supplied.	Is there an approved design/copy source, and what release target and QA/dependency owners should be recorded?
11	Current test readiness	⚠️ Ticket comments record four automated runs, each inconclusive with 0 passed, 0 failed, and 4 blocked; “Actual” is blank.	What specifically blocked execution, and has the blocker been resolved? Please provide the latest reproducible environment/result before sign-off.

3. Test Scenarios

Priorities are proposed test-plan priorities, not a resolution of the ticket’s conflicting priority values. **P0** covers core authentication/security paths; **P1** covers important negative, compatibility, and usability coverage; **P2** covers additional robustness once requirements and environment are available.

ID	Priority	Type	Scenario	Maps to (AC / gap)
TS-01	P0	Positive	Open the login page and verify both exact button labels are visible below “Sign in with Google” in the specified order.	AC: UI Visibility
TS-02	P0	Positive	Select SSO, complete authentication with a valid test identity at each configured provider, and verify an authenticated session and dashboard redirect.	AC: SSO Login Flow, Compatibility; Gap 2
TS-03	P0	Positive	Select Passkey on a supported browser/device, verify with a previously registered passkey, and confirm an authenticated session and dashboard redirect.	AC: Passkey Login Flow, Compatibility
TS-04	P0	Security	Inspect the authentication flow’s browser-visible UI, URL, storage, and network/log outputs to confirm secrets, tokens, and sensitive credentials are not exposed; verify secure protocol use against the approved configuration.	AC: Security; Gap 7
TS-05	P1	Negative	Attempt SSO authentication with a provider-rejected test identity and verify the approved error, safe recovery/retry behavior, and no authenticated session.	AC: SSO Login Flow, Error Handling; Gaps 2–3
TS-06	P1	Negative	Start Passkey login when the account has no registered passkey; verify the approved fallback/guidance and that no session is created.	AC: Passkey Login Flow; Gap 3
TS-07	P1	Negative	On a supported environment, reject/cancel the Passkey prompt or let it time out; verify the agreed message, retry/fallback behavior, and unauthenticated state.	AC: Error Handling; Gap 3
TS-08	P1	Negative	Cause a Passkey verification failure using an approved negative test fixture; verify safe error handling, no login, and no sensitive detail in UI/logs.	AC: Error Handling, Security; Gaps 3, 7
TS-09	P1	Compatibility	Attempt Passkey login on a browser/device confirmed as unsupported; verify the specified unsupported-environment message or guidance and continued access to other login options.	AC: Error Handling, Compatibility; Gap 6
TS-10	P1	Compatibility	Execute the SSO success and rejection paths against every provider in the approved configuration matrix.	AC: Compatibility; Gap 2
TS-11	P1	Regression	Verify existing login methods and their established success/failure behavior remain available and unchanged alongside the new options.	AC: UI Visibility; Gap 9
TS-12	P1	Accessibility	Use keyboard-only navigation and a screen reader to reach, identify, and activate both controls; verify focus order/visibility and error announcement against the agreed accessibility target.	Ticket test scenario; Gap 8
TS-13	P1	Boundary	Resize supported viewport sizes and test mobile/responsive layouts; verify controls remain visible, correctly ordered, operable, and do not obscure existing login methods.	Ticket test scenario; Gap 6

ID	Priority	Type	Scenario	Maps to (AC / gap)
TS-14	P2	Negative	Verify recovery from provider unavailable, malformed/expired SSO response, or interrupted callback when such cases are supported by the test environment; ensure no partial session is created.	AC: Error Handling / secure flow; Gaps 2, 3
TS-15	P2	Boundary	Exercise repeated attempts and refresh/back navigation during an in-progress SSO or Passkey attempt; verify no stale or duplicate authenticated state and the agreed retry behavior.	AC: Error Handling; Gaps 3, 5
TS-16	P2	Non-functional	Measure SSO and Passkey completion/feedback time against an agreed performance target and verify safe, useful diagnostics are available without recording secrets.	Gaps 7–8
TS-17	P2	Localization	If multiple locales are supported, verify button labels, errors, and guidance are translated, fit the UI, and retain correct meaning.	Gap 8

4. Test Data & Environment

- **Data:** To be supplied by the ticket owner/identity team: a successful SSO test identity and provider tenant per configured IdP; a provider-rejected test identity; accounts with and without a registered Passkey; and safe negative fixtures for cancellation, timeout, and verification failure. Avoid using production credentials or real user secrets.
- **Environment / flags:** Use the approved non-production staging environment with the feature enabled and working callback/WebAuthn origins. Exact URL, flag name/value, and deployment build are TBD. The ticket comments refer to `app.vwo.com` and `app.wingify.com/#/login`, with all listed automation cases blocked; they do not establish an approved QA environment.
- **Roles / permissions:** Account/tenant roles and expected authorization behavior are not specified; confirm whether role-specific SSO policies or tenant boundaries need coverage before sign-off.
- **Platform matrix:** Pending an owner-approved list of browsers, versions, operating systems, devices, and authenticators. Include at least the agreed supported and unsupported Passkey environments.
- **Dependencies:** Configured OAuth/SAML identity providers, reachable callback endpoints, WebAuthn-capable browser/device for positive Passkey coverage, existing passkey enrollment (registration itself is out of scope), and test accounts.

5. Risks & Assumptions

- **Assumptions made:**
- The ticket acceptance criteria are the requirements baseline; unspecified behavior is recorded as a question rather than treated as a requirement.
- “Dashboard” means the authenticated user’s expected landing page; its exact route and post-login destination rules require confirmation.
- Passkey tests use accounts enrolled through the separate registration story; registration is not tested here.
- Test scenarios use approved non-production identities and providers.
- **Risks:**
- The issue metadata and description conflict on priority.
- Four recorded automation executions (Oct 10, 13, 17, and 20, 2026) are inconclusive: 0 passed, 0 failed, 4 blocked each time, with blank actual results. The cause is unknown, and the ticket status is To Do; readiness and implementation behavior are therefore unverified.
- Undefined provider coverage, error copy, supported platforms, and test identities can block reproducible execution or produce inconsistent acceptance decisions.
- Broad security wording may leave important token/credential exposure, callback, and logging expectations unverified unless security owners define measurable checks.
- **Company context:** The supplied profile describes VWO as a Wingify product focused on website A/B testing. No additional company or product claims are assumed.

6. Entry / Exit criteria

- **Entry:**
- Build is deployed to an agreed non-production environment, and VWO-49 is available for QA.
- Login page, SSO providers/callbacks, and Passkey origin are configured and reachable.
- Valid/invalid SSO test identities and registered/unregistered Passkey accounts are available.
- Supported browser/device matrix, expected messages/recovery behavior, and priority conflict are clarified.
- The blocker from the ticket's prior inconclusive runs is understood or demonstrably resolved.
- **Exit:**
- All P0 and agreed P1 scenarios are executed with recorded actual results and evidence.
- No open Critical/High defects remain; lower-severity issues have documented disposition and owner approval.
- Security checks show no sensitive authentication data exposure and use the approved protocol configuration.
- Supported and unsupported environment behavior, regression, and accessibility checks meet the agreed criteria.
- Any blocked or deferred scenarios have explicit rationale, owner, and risk acceptance.
- QA owner approves the plan and execution results.

7. Resource Allocation

Role	Allocation / responsibility
QA owner	TBD — approve scope, priority, open questions, and exit decision
QA tester	TBD — execute functional, compatibility, regression, and accessibility scenarios
Identity-provider / authentication engineer	TBD — provide test tenant/accounts and troubleshoot provider/callback dependencies
Security reviewer	TBD — confirm protocol, token-handling, and logging expectations
Product/design owner	TBD — confirm copy, behavior, supported matrix, and priority

8. Timelines, Sprint Details

- **Sprint / fix version:** Not specified in the ticket export.
- **Execution dates / duration:** TBD after environment readiness, test data, and open-question resolution.
- **Milestones:** Resolve requirements and prior-run blocker → prepare test environment/data → execute P0/P1 → execute agreed P2 → QA review and sign-off.

HUMAN REVIEW GATE

- **I assumed:** Testing uses non-production accounts and configured providers; Passkey registration is covered elsewhere; the ticket AC is the source of truth where details are explicit.
- **I could not confirm:** Current implementation/deployment, approved test environment, SSO providers, supported Passkey platform matrix, test data, exact error/fallback copy, performance/accessibility/localization targets, feature flag/rollback, or the cause of prior blocked runs.
- **Open questions blocking sign-off:** Resolve the priority conflict and questions in Section 2, especially provider/platform coverage, failure behavior, test setup, security expectations, and the repeated blocked test executions.
- **Approve or edit this draft before it is treated as approved. Do not start detailed test-case or automation authoring until a human approves the plan.**